

Hardware building guide for an Open-source, High-torque, and Portable (OpenHi-P) hip exoskeleton

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Bill of materials

Commercially available goods

- COM-1: Vest: [Smith Performance Sprayer](#) (\$39.49) x 1
- COM-2: Back brace: [Nvorliy LSO Lower Back Brace](#) (\$38.99) x 1
- COM-3: Hip brace: [Breg T-scope hip brace](#) (used) x 2

Electronics and power

- ELE-1: Motor: [Cubemars AK10-9 V2.0 KV60](#) (\$798.90) x 2
- ELE-2: Load cell: [SMD S230](#) (\$210.94) x 2
- ELE-3: IMU: [Xsens MTi-3-5A-T](#) (\$359) x 2
- ELE-4: Controller: [STM32 Nucleo-F446RE](#) (\$14.85) x 1
- ELE-5: Battery: [MaxAmps LiPo 3250mAh 6S 22.2v Battery Pack](#) (\$210) x 1
- ELE-6: Emergency stop: [Galco XA1E-BV3SG01RH](#) (\$44.93) x 1

Items for fabrication

- FAB-1: Carbon fiber plate: [DragonPlate EconomyPlate™ ~ 1/8" x 24" x 24"](#) (\$337.55)
- FAB-2: Carbon fiber tube: [Rectangle - Spread Tow Fabric - 0.625 x 1.25 - 0.75 x 1.35 x 66 inch](#) (\$206.99)
- FAB-3: Aluminum sheet: [7075-T6 aluminum sheet, 0.16" x 12" x 12"](#) (\$54.9)
- FAB-4: Velcro: Extra-Wide Hook and Loop
 - A. Loop: [VELCRO Brand Sew On Loop, 4" x 900"](#) (\$ 76)
 - B. Hook: [VELCRO Brand Sew On Hook, 4" x 900"](#) (\$ 76)
- FAB-5: Foam Padding: [Adhesive-backed Neoprene Sheet, 12" x 59" x 1/2"](#) (\$26.99)
- FAB-6: Carbon fiber sheet: [Carbon Fiber Fabric 2x2 Twill 6k 11oz/372gsm](#) (\$42.48)
- FAB-7: Epoxy resin: [WEST SYSTEM 105B Bundle with 205B Fast Epoxy Hardener](#) (\$203.22)
- FAB-8: Vacuum bag: [Vacuum Compression Space Saving Storage Bags](#) (\$17.48)
- FAB-9: Peel ply: [ZENFUN Cloth Carbon Fiber Fiberglass 5 Yard 30 Inch](#) (\$15.99)
- FAB-10: Breather-bleeder fabric: [Fibre Glast \(polyester\), ~60" x 36" \(1 yard\)](#) (\$13.95)
- FAB-11: Release film: [Fibre Glast Low Temp Release Film](#) (\$10.95)

Fasteners

Standoff:

- STF-1: [Lightweight Aluminum Female Threaded Hex Standoff](#)

Bolts:

- BLT-1: [Flanged Button Head Screws, M6 x 1.00 mm Thread, 35mm Long](#)
- BLT-2: [Hex Head Screw Flanged, M5 x 0.8 mm Thread Size, 16 mm Long](#)
- BLT-3: [Button Head Hex Drive Screw, Black-Oxide Alloy Steel, M5 x 0.80 mm Thread, 6 mm Long](#)
- BLT-4: [Nylon Pan Head Phillips Screws M3 x 0.50 mm Thread, 25mm](#)
- BLT-5: [Button Head Hex Drive Screw, Black-Oxide Alloy Steel, M6 x 1.00 mm](#)

Thread, 12 mm Long

BLT-6: Black-Oxide Alloy Steel Hex Drive Flat Head Screw 8-32 Thread Size, 3/4" Long

BLT-7: Black-Oxide Alloy Steel Socket Head Screw 10-32 Thread Size, 3/4" Long

BLT-8: Zinc-Plated Alloy Steel Hex Drive Flat Head Screw M4 x 0.7 mm Thread Size, 10 mm Long

BLT-9: Button Head Hex Drive Screw Black-Oxide Alloy Steel, M4 x 0.70 mm Thread, 10 mm Long

BLT-10: Black-Phosphate Class 10.9 Steel Hex Head Screw Flanged, M5 x 0.8 mm Thread Size, 25 mm Long

Nuts:

NUT-1: Zinc-Plated Steel M5 x 0.8 mm

NUT-2: Nylon Hex Nut M3 x 0.5 mm Thread

Washer:

WSR-1: Stainless Steel Oversized Washer, 5.3 mm ID, 15.0 mm OD

WSR-2: Stainless Steel Oversized Washer, 6.4 mm ID, 18.0 mm OD

WSR-3: Stainless Steel Oversized Washer 0.172" ID, 0.375" OD

Fabrication methods

Carbon-fiber composite molding

Components:

- CF-1: *Thigh_strut_mold_left.stl*, *Thigh_strut_mold_right.stl*
- CF-2: *Waist_c_frame_mold.stl*
- CF-3: COM-2 reinforcement

Steps:

1. Print molds (*Thigh_strut_mold_left.stl*, *Thigh_strut_mold_right.stl*, and *Waist_c_frame_mold.stl*). Cover all mold surfaces with release film to prevent epoxy adhesion and to facilitate demolding after curing.
2. Cut the carbon fiber sheets, peel ply, and breather fabric to the required geometries. Apply two layers of peel ply to CF-1 and CF-2 to ensure clean release and a consistent surface finish. For CF-3, apply one layer of peel ply. In all cases, apply a single layer of breather fabric to enable air evacuation and absorb excess resin during vacuum compaction.
3. Prepare the epoxy using WEST SYSTEM 105B resin with 205B fast hardener. Mix the components according to the manufacturer's recommended ratio and stir thoroughly until the mixture is homogeneous to ensure consistent curing. Perform all layups in a chemical fume hood under vacuum bagging conditions to ensure proper resin handling and adequate laminate compaction.
4. For CF-1 and CF-2, first place a layer of peel ply onto the prepared mold surface, then apply a thin, uniform epoxy layer. Use 8 carbon fiber layers for CF-1, 16 layers for CF-2. For CF-3, apply epoxy directly to the central support region of the COM-2 brace, then bond two carbon fiber sheets to both the anterior and posterior surfaces. Use 4 layers for each side of CF-3.
5. Apply carbon fiber sheets sequentially, brushing epoxy between layers to ensure complete saturation and adequate interlaminar bonding. Use a balanced $0^{\circ}/90^{\circ}/\pm 45^{\circ}$ fiber orientation sequence to provide multidirectional stiffness and improved structural performance under combined loading. After placing the final carbon fiber layer, apply a top layer of peel ply to facilitate unmolding.
6. Cover the laminate stack with breather fabric and enclose it in vacuum bagging film. Seal the vacuum bag and apply vacuum pressure to ensure uniform compaction across the laminate thickness. Inspect the system for leaks and confirm that the bag conforms tightly to the mold geometry. Cure the composite under continuous vacuum at room temperature overnight.
7. Once curing is complete, release the vacuum and remove the composite parts from the bag. Remove the breather fabric, peel ply, and release film, then trim excess material from the edges. Achieve the final edge geometry by grinding and hand-drilling mounting holes according to specification.

Waterjet cutting

Components:

- Carbon fiber plate (FAB-1, 0.125 inch)
 - CUT-CF-1: *Waist_back_plate.dwg* x 2
 - CUT-CF-2: *Waist_side_plate.dwg* x 2
- Aluminum plate (FAB-3, 0.16 inch)
 - CUT-AL-1: *Motor_rotor.dwg* x 2
 - CUT-AL-2: *Motor_stator.dwg* x 2
 - CUT-AL-3: *Motor_stator_strut.dwg* x 2
 - CUT-AL-4: *Loadcell_bumper.dwg* x 4

CNC machining

Components:

- Aluminum 7075
 - CNC-1: *Insert.stl* x 4
 - After printing, press-fit two NUT-1s into the designated hexagonal recesses.

3D-printing

Components:

- 3DP-1: *Waist_side_connector.stl* x 2
- After printing, press-fit two NUT-1s into the designated hexagonal recesses.

Settings:

- Printer: Bambu Lab P1S
- Setup: 0.4 mm nozzle, textured PEI build plate, Infill 100%.
- Print material: Carbon fiber-reinforced PLA (PLA-CF).

Sewing

Components:

- COM-1

Steps:

1. Remove the waist parts of COM-1. Sew nylon webbing horizontally along both sides of COM-1 to create reinforced attachment points. Secure zip ties around the back rod and the webbing bands to minimize spacing and restrict movement of the carbon fiber rod.
2. Sew an additional strip of nylon webbing along the posterior centerline to guide and stabilize the carbon fiber rod. Reuse the trimmed webbing from the initial fabric removal step for this purpose. Maintain a narrow webbing channel profile to ensure a tight fit with the carbon fiber rod.
3. Sew Velcro to the bottom of COM-1 to complete the lumbar support interface.

Manual machining

Components:

- Carbon fiber tube cutting (FAB-2)
 - CUT-CF-3
 - Cut FAB-2 to 55 cm.
 - Drill two through-holes using the reference model (*Cf_tube_hole_reference.stl*)
- Load cell tapping (ELE-2)
 - Tap two holes for an 8-32 thread (0.15 inch diameter).
 - Tap one hole for a 10-32 thread (0.17 inch diameter).
- Countersinking (CUT-AL-2)
 - Countersink the flat-head bolt holes.
- COM-2 post-processing
 - Drill two M6 holes along the centerline of the brace, spaced 85 mm apart, with the lower hole located 95 mm from the bottom edge.
 - Attach the Velcro to the lumbar support side of COM-2.

Assembly steps

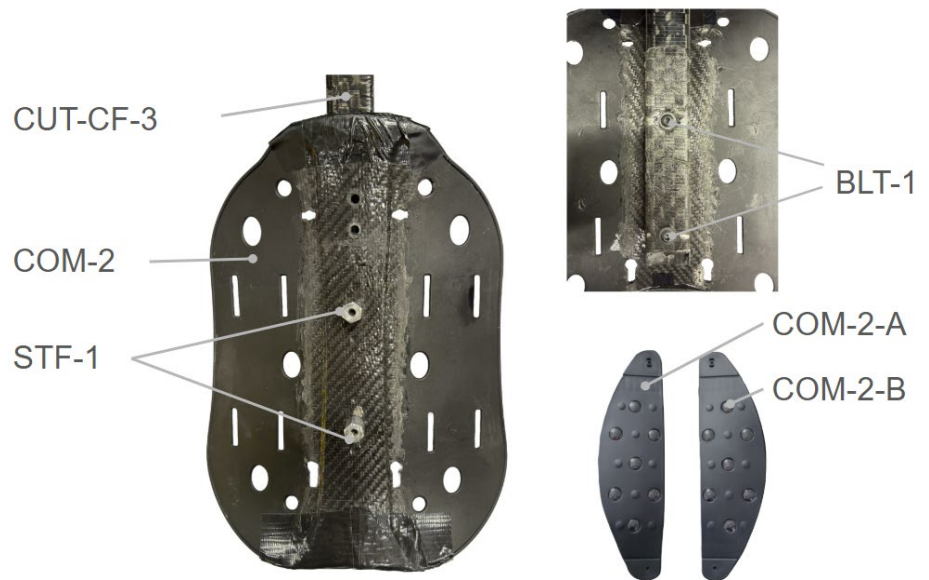
A1. Back brace and strut integration

Parts needed:

- CUT-CF-3
- COM-2
- BLT-1 x 2
- STF-1 x 2

Steps:

1. Secure CUT-CF-3 to COM-2 using two BLT-1s and two STF-1s.
2. Trim FAB-4-A to match the geometry of COM-2-A and COM-2-B.
3. Bond FAB-4-A to COM-2 to provide a compliant contact surface and improve load distribution at the lumbar interface.



A2. A1 and vest integration

Parts needed:

- A1
- COM-1

Steps:

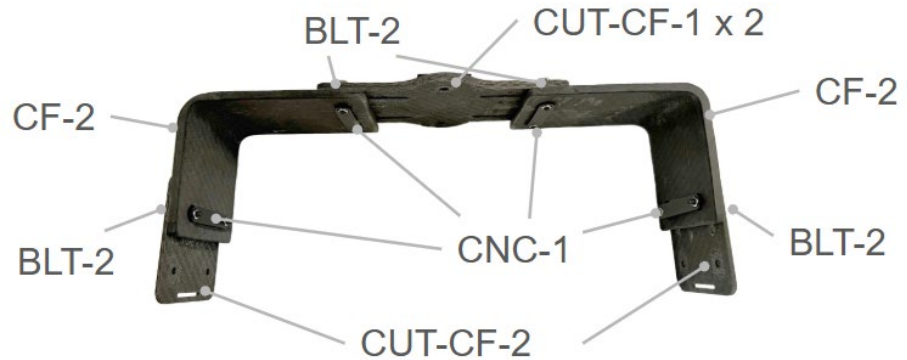
1. Insert CUT-CF-3 (A1) into COM-1.
2. Adjust its vertical position to match the wearer's torso length to ensure proper fit and comfort.
3. Thread the COM-1 shoulder strap webbing through the outer slot on COM-2 (A1) to secure the brace assembly to the vest and maintain alignment during wear.



B1. Carbon fiber parts integration

Parts needed:

- CUT-CF-1 x 2
- CUT-CF-2 x 2
- CF-2 x 2
- CNC-1 x 4
- WSR-1 x 8
- BLT-2 x 8



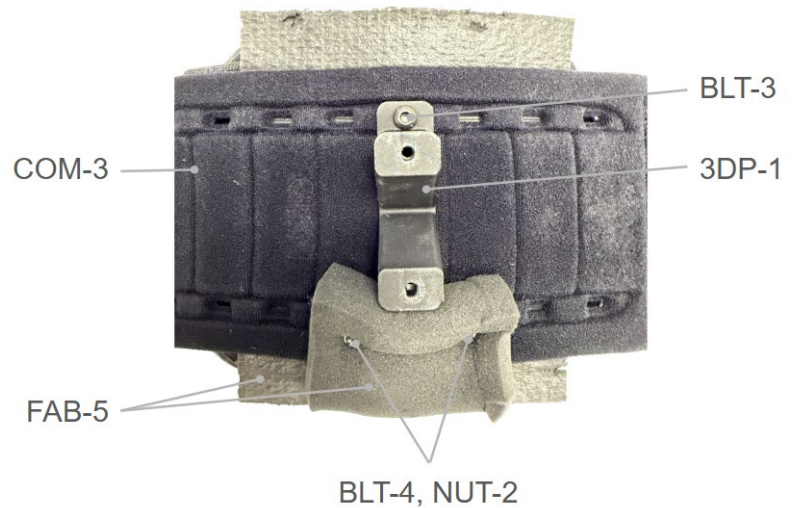
Steps:

1. Install the CNC-1 inserts into the corresponding CF-2 components to create reinforced threaded interfaces.
2. Fully overlap and align the two CUT-CF-1 plates to form the central plate structure.
3. Secure each side of this assembly to the CF-2 plates using BLT-2 and WSR-1.
4. Attach the two CUT-CF-2 plates to the opposite sides of the CF-2 plates using the remaining BLT-2 and WSR-1 to complete the carbon fiber plate subassembly.

C1. Soft waist cover integration

Parts needed:

- COM-3 x 1
- FAB-5
- 3DP-1 x 2
- BLT-3 x 4
- BLT-4 x 4
- NUT-2 x 4



Steps:

1. Secure 3DP-1 to the cavity of COM-3 using BLT-3.
2. Trim FAB-5 and wrap it around the edge and lower portion of COM-3 to cover both the

- inner and outer surfaces and to improve pressure distribution during wear.
3. Secure the wrapped FAB-5 layer at the bottom holes using BLT-4 and NUT-2.

C2. A2, B1, and C1 integration

Parts needed:

- A2
- C1
- B1
- BLT-5 x 2
- WSR-2 x 2



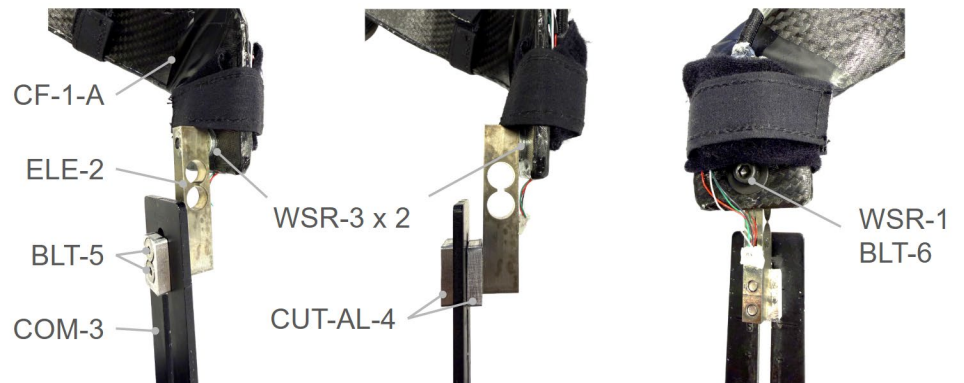
Steps:

1. Position C1 on top of A2. Ensure that the STF-1 does not interfere with or compress the routed wires during installation.
2. Secure CUT-CF-1 and STF-1 using BLT-5 and WSR-2.

D1. Thigh parts integration

Parts needed:

- CF-1-A
- CF-1-B
- COM-3
- ELE-2 x 2
- CUT-AL-4 x 4
- BLT-5 x 4
- BLT-6 x 2
- WSR-1 x 2
- WSR-3 x 4



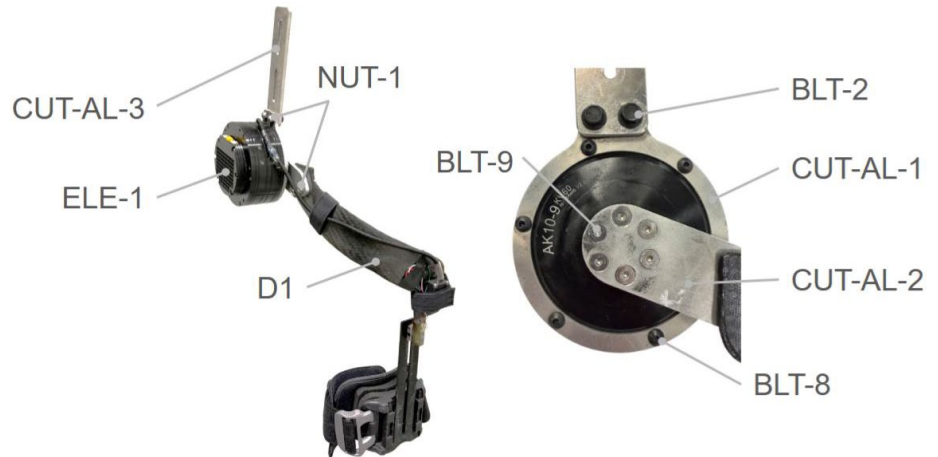
Steps:

1. Stack two WSR-3 in sequence on the side of CF-1-A.
2. Insert BLT-6 through both WSR-3 and CF-1-A, then tighten it into the thread of ELE-2.

D2. D1 and motor integration

Parts needed:

- D1
- CUT-AL-1 x 2
- CUT-AL-2 x 2
- CUT-AL-3 x 2
- ELE-1 x 2
- BLT-2 x 8
- BLT-8 x 6
- BLT-9 x 6
- NUT-1 x 8



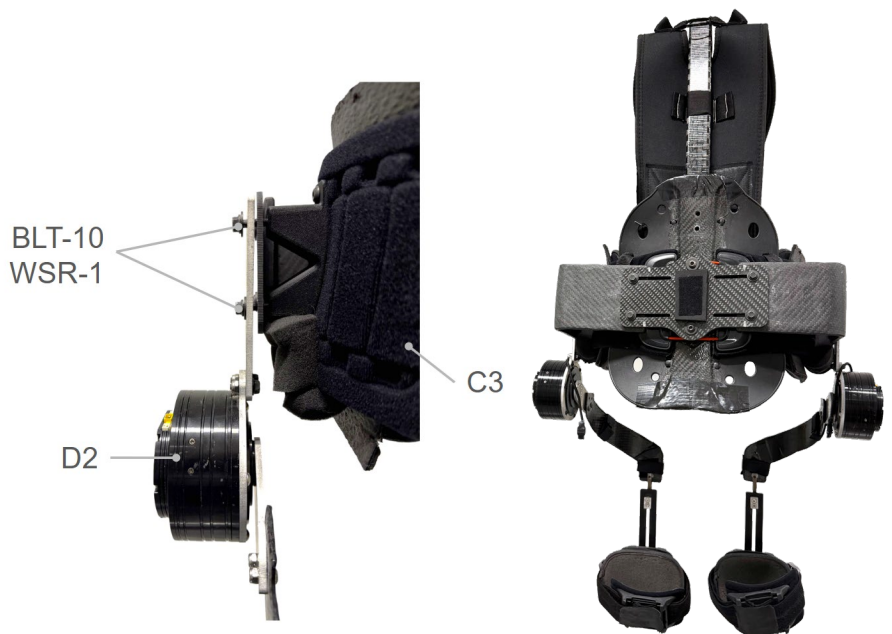
Steps:

1. Secure CUT-AL-1 to ELE-1 using BLT-8 bolts.
2. Attach CUT-AL-2 to the output shaft of ELE-1 using BLT-9 bolts to ensure proper mechanical coupling.
3. Fasten CUT-AL-3 to CUT-AL-1 using BLT-2 and NUT-1 to form the motor mounting linkage.
4. Secure CUT-AL-2 to CF-1-B of D1 using BLT-2 and NUT-1 to complete integration of the motor assembly with the thigh structure.

E1. C2 and D2 integration

Parts needed:

- C2
- D2
- BLT-10 x 4
- WSR-1 x 4



Steps:

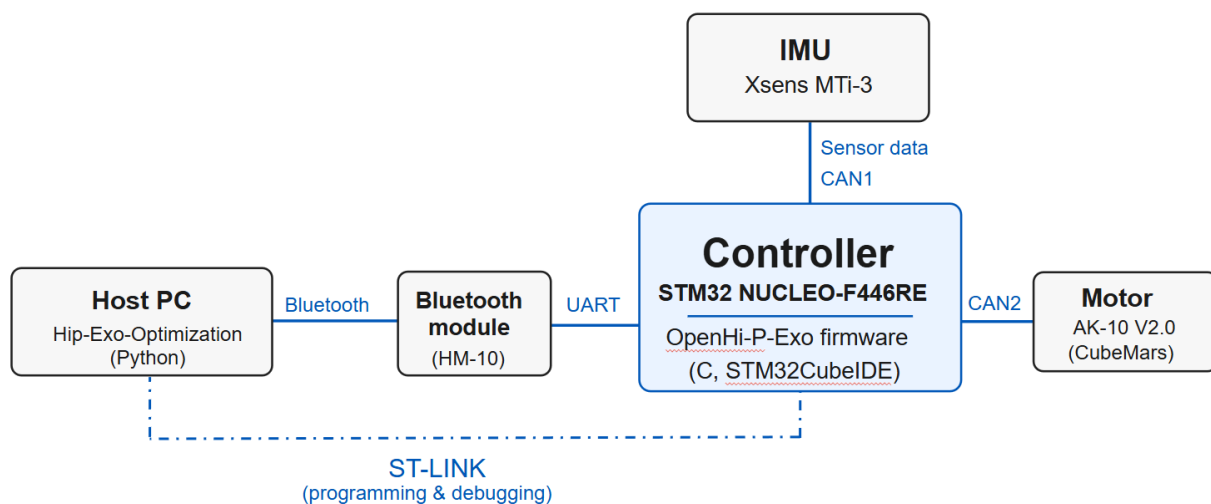
1. Align CUT-AL-3 and CUT-CF-2 with 3DP-1.
2. Secure the components together using BLT-10 and WSR-1 to complete integration of the C2 and D2 assemblies and finalize the hardware assembly.

Electronics

The OpenHi-P control system is divided into a host-computer layer and an embedded real-time layer.

1. The host PC runs the human-in-the-loop optimization software in the Optimizer directory of the OpenHi-P-Exo repository.
2. The embedded controller runs the C firmware in the Controller directory of the OpenHi-P-Exo repository.

These two software systems communicate through the HM-10 Bluetooth module.



During an optimization condition:

1. The host PC generates a candidate hip-torque profile and sends the profile over Bluetooth Low Energy.
2. The HM-10 converts the wireless data to a 3.3 V UART stream.
3. The STM32 receives the packet through USART1, updates the active torque profile, estimates gait phase from the sensor signals, and sends motor-current commands through CAN2.

The embedded controller is the STM32F446RE on the NUCLEO-F446RE development board. The firmware project was created for STM32CubeIDE 1.17.0 and uses the STM32 HAL and FatFs libraries included in the Controller directory. The system clock is configured at 168 MHz.

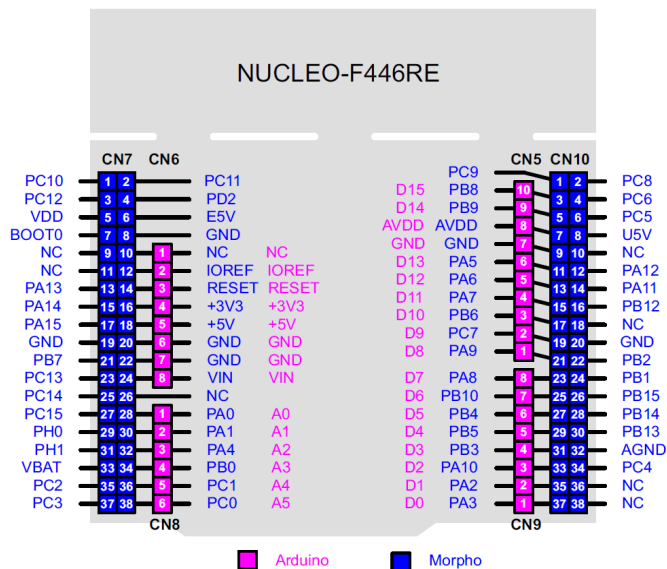


- 1. Clone the repository.** Clone OpenHi-P-Exo and open the Controller directory.
- 2. Import the project.** In STM32CubeIDE, select File > Import > Existing Projects into Workspace and choose the Controller directory.
- 3. Build.** Select the Debug or Release configuration and build the project.

4. Connect ST-LINK. Connect the NUCLEO-F446RE to the PC through the onboard ST-LINK USB connector.

5. Flash and test. Use Run or Debug to program the MCU. Confirm that the firmware starts without build or connection errors.

The peripheral configuration is stored in Open Hip Exo.ioc. If STM32CubeMX is used to regenerate source files, application changes should remain inside the generated USER CODE regions.



IMU and CAN interface

The OpenHi-P system uses an Xsens MTi-3 attitude and heading reference system on each leg. The sensor combines a three-axis gyroscope, accelerometer, and magnetometer with onboard sensor fusion. It provides orientation (roll, pitch, and yaw), linear acceleration, and angular-rate measurements. Left-leg frames use the 0x300 block, right-leg the 0x500 block.

Signal	Left-leg CAN IDs	Right-leg CAN IDs	Values stored by the STM32
Orientation	0x311-0x312	0x511-0x512	Roll, pitch, yaw
Acceleration	0x313-0x314	0x513-0x514	AccX, AccY, AccZ
Angular rate	0x315-0x316	0x515-0x516	GyroX, GyroY, GyroZ

CAN ID (left / right)	Bytes 0-3	Bytes 4-7	Encoding
0x311 / 0x511	Roll	Pitch	float32, big-endian
0x312 / 0x512	Yaw	-	float32, big-endian
0x313 / 0x513	AccX	AccY	float32, big-endian
0x314 / 0x514	AccZ	-	float32, big-endian
0x315 / 0x515	GyroX	GyroY	float32, big-endian
0x316 / 0x516	GyroZ	-	float32, big-endian

Datasheet and integration references: [Xsens MTi-3 product sheet](#); [Xsens MTi 1-series hardware documentation](#).

Actuator and CAN interface

Each hip joint is driven by a CubeMars AK10-9 V2.0 actuator. The actuator integrates a brushless motor, a 9:1 planetary gearbox, position sensing, and a motor driver. The embedded driver closes the inner motor-current loop, while the STM32 calculates the current command needed to produce the desired hip torque.

Command and feedback flow between the STM32 and the motor:

- 1. Generate desired torque.** The gait controller evaluates the active torque profile for the current gait phase.
- 2. Convert torque to current.** The current controller uses the torque constant and gear ratio, then adds torque-error feedback and motor-speed damping. The sign is adjusted for the left and right actuators.
- 3. Send the CAN2 command.** At up to 500 Hz, the STM32 sends an extended CAN frame to motor ID 0 (right) or motor ID 1 (left). The four-byte payload stores current in amperes multiplied by 1000.
- 4. Read motor feedback.** The actuator returns position, electrical speed, current, temperature, and an error code. The STM32 uses these values for control, logging, and safety checks.

Item	OpenHi-P configuration
Power	22.2 V nominal actuator supply from the 6S battery pack (25.2 V full charge); the AK 2.0 driver manual specifies an 18-52 V input range.
CAN connection	CAN2 on PB12 (RX) and PB13 (TX), through the CAN transceiver; 1 Mbit/s; extended frames.
Current command	Extended ID = motor_id (1 << 8); four-byte signed current command, current [A] x 1000.
Periodic feedback	Extended ID = 0x2900 + motor_id; eight bytes for position, speed, current, temperature, and error code.
Motor IDs	0 = right actuator; 1 = left actuator.

Before enabling assistance, configure unique motor IDs, select servo mode, set the CAN rate to match the controller, and enable periodic status feedback. The firmware sends zero current when control mode is off or gait has not been detected. It also stops assistance when the configured current, torque, or temperature limits are exceeded.

Datasheet and setup references: [CubeMars AK10-9 V2.0 product page](#); [AK 2.0 series driver manual](#).

Power distribution and grounding

The exoskeleton runs from a single lithium-polymer battery pack (ELE-5). The pack feeds two branches: an unregulated actuator bus and a regulated logic bus.

- 1. Actuator bus.** Both actuators are supplied directly from the pack: 22.2 V nominal, 25.2 V full charge (ELE-5, 6S LiPo). The AK 2.0 driver accepts 18-52 V. Set the low-voltage cutoff at 3.2 V per cell (19.2 V), since 3.0 V per cell reaches the 18 V driver minimum.
- 2. Logic bus.** A buck regulator converts the pack voltage to 5 V for the NUCLEO-F446RE. It must accept the full pack range and supply at least 1 A. The build uses a Pololu D24V22F5 (6.5-36 V in, 5 V 2.5 A); any equivalent may be substituted.
- 3. Board and sensor rails.** The 5 V enters the Nucleo at E5V (CN7 pin 6; ground CN7 pin 8). Set JP5 to position 2-3 and leave JP1 open. The onboard regulator then supplies 3.3 V to the HM-10, the sensor interface, and the MTi-3 sensors. ST-LINK USB may stay connected while powered from E5V.
- 4. Grounding.** Tie pack negative, regulator ground, Nucleo GND, CAN transceiver grounds, and HM-10 ground to a single common ground. Both CAN buses and the HM-10 UART link require it.

Rail	Source	Nominal voltage	Loads
Actuator bus	Battery pack (ELE-5)	22.2 V nominal (25.2 V full charge; driver range 18-52 V)	AK10-9 V2.0 actuators (x2)
Logic bus	Buck regulator from the pack	5 V	NUCLEO-F446RE, through E5V
Logic 3.3 V	Nucleo onboard regulator	3.3 V	HM-10, MTi-3, sensor interface, CAN transceivers

Bluetooth module setup



The HM-10 module provides a wireless serial link between the host PC and USART1 on the STM32. The module contains its own BLE firmware and acts only as a BLE-to-UART bridge.

The following items are required to configure and test the HM-10 module:

- HM-10 BLE module
- USB-to-UART converter
- Jumper wires

The USB-to-UART converter is used only to configure and test the HM-10 module. During exoskeleton operation, the HM-10 is connected directly to the STM32 controller.

Connect the module to a PC through a 3.3 V USB-to-UART converter. Begin at the module default baud rate, which is commonly 9600 baud, and enter AT-command mode. First send AT and confirm an OK response. Then configure the BLE name and UART baud rate so they match the host optimizer and STM32 firmware.

```
AT
AT+NAMEHIP_EXO_MAIN
AT+BAUD4
```

The module should return an OK response after the initial AT command. The remaining commands set the BLE device name to HIP_EXO_MAIN and configure the UART baud rate to 115200 baud. After changing the baud rate, reconnect the serial terminal at 115200 baud and send AT again to confirm that communication still works.

Wiring the HM-10 to the STM32

After configuration, connect the HM-10 to the NUCLEO-F446RE as shown below.

HM-10 pin	STM32 connection
VCC	3.3 V
GND	GND
TXD	PB7 / USART1_RX
RXD	PB6 / USART1_TX

Both devices must share the same ground. The HM-10 used in this system operates with 3.3 V logic and should be connected to the 3.3 V supply of the controller.